

Inputs

- Observed unphased genotypes: Dad = 0/1, Mom = 0/1, Kid1 = 0/1, Kid2 = 0/1, Kid3 = 1/1.
- Inferred founder haplotype each kid carries: Kid1 = A|C, Kid2 = B|D, Kid3 = A|D.

Algorithm

- Enumerate the 4 ways to assign alleles (0 and 1) to dad's two haplotypes A, B and mom's two haplotypes C, D consistent with the parents' unphased genotypes. For each such assignment:
1. derive each kid's expected phased genotype by reading off the alleles assigned to the two founder haplotypes it carries (e.g., A|C → dad's A allele assignment paired with mom's C allele assignment);
 2. for each kid, unphase the expected genotype and tick (✓) if it matches the kid's observed unphased genotype (from Inputs);
 3. count kids whose expected unphased genotype agrees with observed (# Matches).

Hap-allele assignment				Expected kid genotypes						
Dad (0/1)		Mom (0/1)		Kid1		Kid2		Kid3		# Matches
A	B	C	D	A C		B D		A D		
0	1	0	1	0 0	0 / 0	1 1	1 / 1	0 1	0 / 1	0
0	1	1	0	0 1	0 / 1 ✓	1 0	0 / 1 ✓	0 0	0 / 0	2
1	0	0	1	1 0	0 / 1 ✓	0 1	0 / 1 ✓	1 1	1 / 1 ✓	3
1	0	1	0	1 1	1 / 1	0 0	0 / 0	1 0	0 / 1	0

Output

The assignment with the most matches (boxed) gives the deduced phased genotypes:
Kid1 = 1|0, Kid2 = 0|1, Kid3 = 1|1.